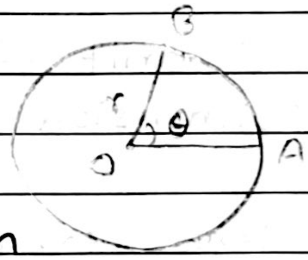
Circular motion

→ If a particle moves in a circular path with uniform speed then the motion is called circular motion.

- Angular displacement (θ)

→ The angular displacement of a particle moving in a circular path is defined as the angle turned out by the radius vector when particle moves from one point to another.



In the figure (θ) represents angular displacement of a particle when it moves from A to B.

The SI unit of Angular displacement (θ) is radian (rad) and it is dimensionless quantity.
i.e. $[M^0 L^0 T^0]$

- Angular Velocity (ω)

→ The rate of change of angular displacement with respect to time is called Angular velocity

$$\therefore \omega = \frac{d\theta}{dt}$$

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SI unit of ω = rad s^{-1}
Dimension of ω = $[M^0 L^0 T^{-1}]$

- Angular Acceleration (α)

↪ The rate of change of angular velocity with respect to time is called Angular Acceleration (α)

$$\therefore \alpha = \frac{d\omega}{dt} = \frac{\omega - \omega_0}{t}$$

SI unit of α = rad s^{-2}
Dimension of α = $[M^0 L^0 T^{-2}]$

- Time period (T)

↪ The time taken by the particle to complete one revolution is called Time period (T).

- frequency (f)

↪ The number of revolution made by the particle in one second is called frequency.

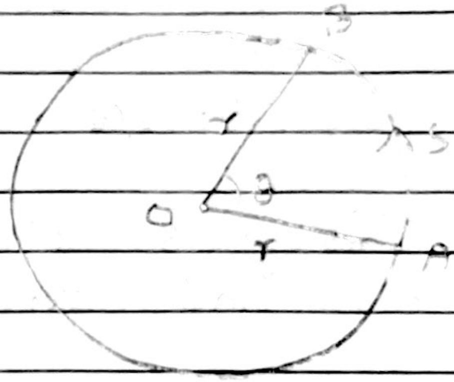
↪ It is given by the reciprocal of time period.
i.e. $f = \frac{1}{T}$

SI unit of f = Hertz or rev/s

Dimension of frequency = $[T^{-1}]$

- Relation between linear and angular parameters:

- Let θ be the angular displacement and s be the linear displacement of a particles when it moves from A to B in a circle of radius r as shown in figure.



From geometry, we have.

$$\theta = \frac{\text{Arc } AB}{\text{radius}}$$

$$\therefore \theta = \frac{\text{Arc } AB}{r}$$

For small θ, Arc AB = s (linear displacement)

So, $\theta = \frac{s}{r}$

$$\therefore s = r\theta \longrightarrow (i)$$

This gives relation between linear displacement and angular displacement.

Differentiating eqn (i) with respect to time, we have.

$$\frac{ds}{dt} = \frac{d(r\theta)}{dt}$$

or, $\frac{ds}{dt} = r\frac{d\theta}{dt}$

or, $v = r\omega \longrightarrow (ii)$

This gives relation between linear velocity and angular velocity.

Again, differentiating eqn (ii) with respect to time, we get.

$$\frac{dv}{dt} = \frac{d(r\omega)}{dt}$$

$$a = r\frac{d\omega}{dt}$$

$$a = r\alpha$$

$$\therefore a = r\alpha \longrightarrow (iii)$$

This gives relation between linear acceleration and angular acceleration.

- Relation between ω, T and F.

↳ Since angular velocity = $\frac{\text{Angular displacement}}{\text{time}}$

$$\therefore \omega = \frac{\theta}{t} \longrightarrow (i)$$

For 1 revolution, we have

$$\theta = 2\pi \text{ radian}$$

and $t = T$ (Time period)

Then from (i)

$$\omega = \frac{2\pi}{T} \rightarrow (2)$$

Since, $f = \frac{1}{T}$

$$\therefore \omega = 2\pi f \rightarrow (3)$$

- Centripetal acceleration and centripetal force.

↪ When a particle moves in a circular path with uniform speed then a force on the particle acts towards the center of the circular path. This force is called Centripetal force.

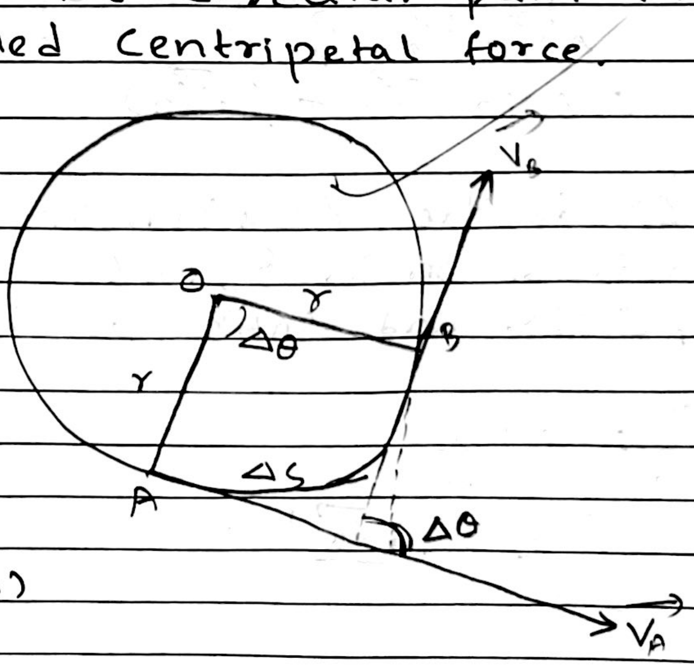


fig (1)

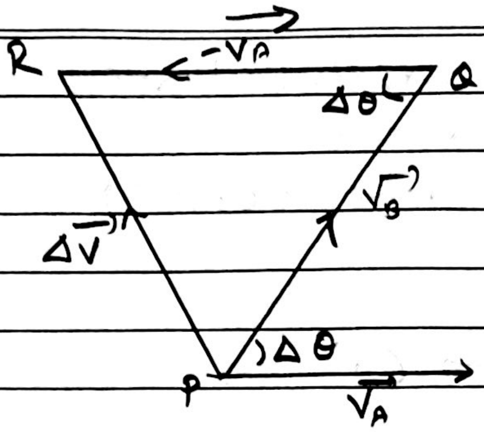


fig (2)

Consider a particle is moving in a circle of radius (r) with uniform speed (V). Let $\Delta\theta$ be the angular displacement and ΔS be the ~~ang~~ linear displacement of the particle when it moves from A to B. in time Δt.

Let, $\vec{V_A}$ and $\vec{V_B}$ be the velocities of the particles at positions A and B then $|\vec{V_A}| = |\vec{V_B}| = V$ And change in velocity of the particle is given by

$$\vec{\Delta V} = \vec{V_B} - \vec{V_A}$$

$$\therefore \Delta V = \vec{V_B} + (-\vec{V_A})$$

Here,

ΔV is the Resultant of $\vec{V_B}$ and $(-\vec{V_A})$ which is shown in figure (2)

Here, ΔPQR and ΔOAB are similar triangles

so,

$$\frac{PR}{AB} = \frac{PQ}{OB}$$

or, $$\frac{|\vec{\Delta V}|}{\Delta S} = \frac{|\vec{V_B}|}{r}$$

or, $\frac{\Delta v}{\Delta s} = \frac{v}{r}$

$\therefore \Delta v = \frac{v}{r} \Delta s \rightarrow$ (i)

or, $\frac{\Delta v}{\Delta t} = \frac{v}{r} \frac{\Delta s}{\Delta t} \Rightarrow \frac{dv}{dt} = \frac{v}{r} \frac{ds}{dt}$

or, $a = \frac{v}{r} \times v$

$\therefore a = \frac{v^2}{r} \rightarrow$ (ii)

The direction of this acceleration is towards the centre of the circle. So, it gives centripetal acceleration.

Since, $v = \omega r$

$\therefore a = \frac{(\omega r)^2}{r}$

$\therefore a = \omega^2 r \rightarrow$ (iii)

This is another form of centripetal acceleration.

If 'm' be the mass of the particle then,

$F = ma$

$\boxed{\therefore F = \frac{mv^2}{r} = m\omega^2 r}$

This is required expression of centripetal force.

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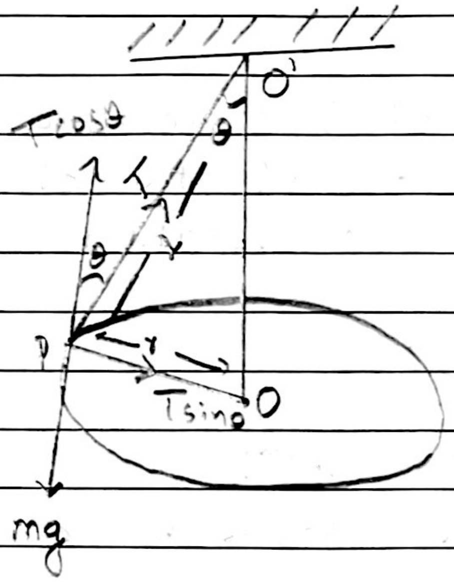
- Motion of particle in a horizontal circle

OR

Conical pendulum

↳ A conical pendulum is a heavy particle suspended from a rigid support by an inextensible, flexible and weightless string which is whirled in a horizontal circle with uniform speed.



Consider a particle of mass 'm' tied at one end of an inextensible string is whirled in a horizontal circle of radius 'r' with uniform speed (v). let 'l' be the length of the string.

The different forces acting on the particle are

(i) its weight (w) = mg, acting vertically downward.

(ii) Tension (T), acting towards the point of suspension, ~~E~~ O'.

The tension (T) can be resolved into two rectangular components $T\sin\theta$ and $T\cos\theta$. as shown in figure.

The component $T\sin\theta$ acts towards the centre of the circle. so, it provides centripetal force to the particle.

$$\therefore T\sin\theta = \frac{mv^2}{r} \longrightarrow (i)$$

And the component $T\cos\theta$ balance the weight of the particles.

$$\therefore T\cos\theta = mg \longrightarrow (ii)$$

dividing (i) by (ii) we have.

$$\frac{\cancel{T}\sin\theta}{\cancel{T}\cos\theta} = \frac{\frac{\cancel{m}v^2}{r}}{\cancel{m}g}$$

or, $\tan\theta = \frac{v^2}{rg}$

or, $v^2 = rg\tan\theta$

or, $v = \sqrt{rg\tan\theta}$

or, $\omega r = \sqrt{rg\tan\theta}$

or, $\omega = \sqrt{\frac{g\tan\theta}{r}}$

or, $\frac{2\pi}{t} = \sqrt{\frac{g\tan\theta}{r}}$ $t \rightarrow$ Time period.

$\therefore t = 2\pi\sqrt{\frac{r}{g\tan\theta}} \longrightarrow$ (iii)

From figure,

$\sin\theta = \frac{OP}{O'P} = \frac{r}{l}$

$\therefore r = l\sin\theta$

then From eqn (iii) we have,

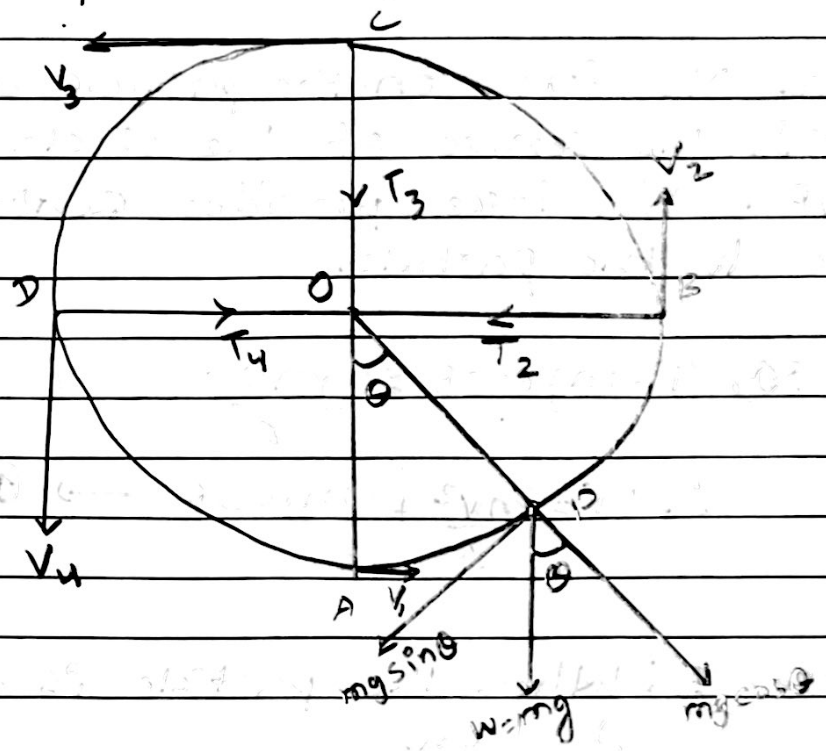
$t = 2\pi\sqrt{\frac{l\sin\theta}{g\tan\theta}}$

or, $t = 2\pi\sqrt{\frac{l\cancel{\sin\theta}}{g\cancel{\sin\theta}} \times \cos\theta}$

$$\boxed{\therefore t = 2\pi\sqrt{\frac{l\cos\theta}{g}}}$$

This gives time period of conical pendulum.

• Motion of particle in a vertical circle



Consider a particle of mass 'm' tied at one end of an inextensible string is whirled in a verticle circle of radius 'r'. Let 'p' be any position of the particle.

The different forces acting on the particle are :-

(i) its weight (w) = mg, acting vertically downward.

(ii) Tension (T), acting towards the centre of the circle.

The weight (mg) can be resolved into two rectangular components mg sinθ and

$mg\cos\theta$ as shown in figure.

Here Net force on the particle acting towards the centre of the circle is $T - mg\cos\theta$. This force provides centripetal force to the particle.

$$So,\ T - mg\cos\theta = \frac{mv^2}{r}$$

$$\therefore T = \frac{mv^2}{r} + mg\cos\theta \longrightarrow (1)$$

<u>Case-I</u>: When the particle is at position A then.

eqn can be written as :-

$$T_1 = \frac{m v_1^2}{r} + mg\cos 0^\circ$$

$$\therefore T_1 = \frac{mv_1^2}{r} + mg$$

This is maximum Tension

$$\boxed{\therefore T_{max} = \frac{mv_1^2}{r} + mg} \longrightarrow (2)$$

<u>Case-II</u>: When the particle is at B then.

$$T_2 = \frac{mv_2^2}{r} + mg\ \cancel{\sin}\ \cos 90^\circ$$

$$\boxed{\therefore T_2 = \frac{mv_2^2}{r}} \longrightarrow (3)$$

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Case III : When the particle is at C then,

$$T_3 = \frac{mv_3^2}{r} + mg\cos 180^\circ$$

$$\boxed{\therefore T_3 = \frac{mv_3^2}{r} - mg} \longrightarrow$$

This is mimimum Tension

i.e.

$$\boxed{\therefore T_{min} = \frac{mv_3^2}{r} - mg} \longrightarrow (4)$$

Case IV : When the particle is at D then,

$$T_4 = \frac{mv_4^2}{r} + mg\cos 270^\circ$$

$$\boxed{\therefore T_4 = \frac{mv_4^2}{r}} \longrightarrow (5)$$

- Minimum velocity of particle at lowest and topmost position of verticle circle.

> At topmost position C, we have.

$$T_3 = \frac{mv_3^2}{r} - mg$$

$$\therefore \frac{mv_3^2}{r} = mg + T_3$$

For the minimum value of V_3 we should have $T_3 = 0$ so,

$$\frac{\cancel{m}V_3^2}{r} = \cancel{m}g$$

or, $V_3^2 = rg$

$\therefore V_3 = \sqrt{rg} \rightarrow 6$

Again,

From principle of conservation of ~~tin~~ ~~ear~~ energy, we have

Total energy at A = Total energy at C

or, P.E at A + K.E at A = P.E at C + K.E at C

or, $mg \times 0 + \frac{1}{2}mv_1^2 = mg \times 2r + \frac{1}{2}mV_3^2$

or, $\frac{1}{2}mv_1^2 = 2mgr + \frac{1}{2}m(\sqrt{rg})^2$

or, $\frac{1}{2}mv_1^2 = 2mgr + \frac{mrg}{2}$

or, $\frac{1}{\cancel{2}}\cancel{m}v_1^2 = \frac{5}{\cancel{2}}\cancel{m}gr$

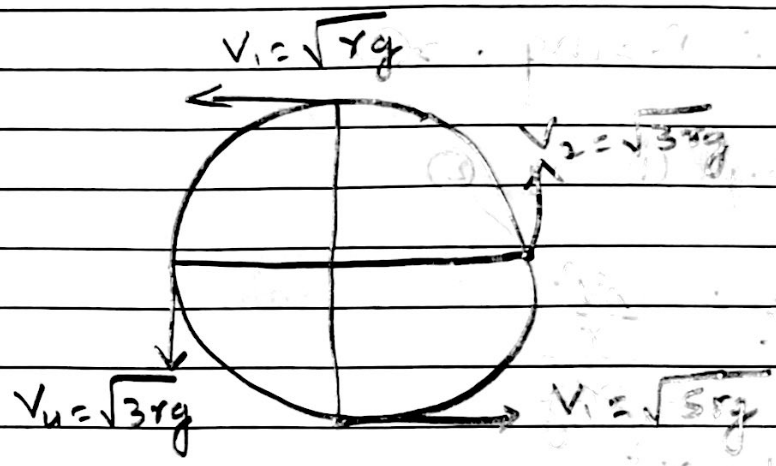
or, $V_1^2 = 5gr$

$\therefore V_1 = \sqrt{5rg} = \sqrt{5}\,V_3 \rightarrow 7$

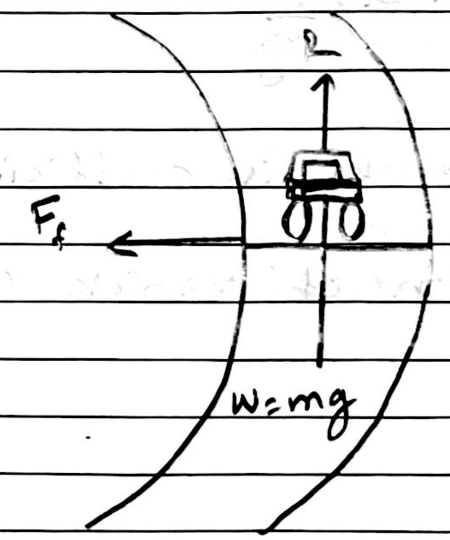
From eqn ⑥ and ⑦ we have,

Velocity of particle at lowest position $= \sqrt{5} \times$ velocity of particle at topmost position.

Note:



- Motion of vehicle on level circular road



Here, force of friction between the road and tyres or vehicles provides centripetal force to the vehicle.

$$\therefore F_f = \frac{mv^2}{r} \rightarrow (1)$$

And,
Weight of the vehicle is balanced by Normal ~~reac~~ reaction.

$$\therefore R = mg \rightarrow (2)$$

Dividing (1) by (2)

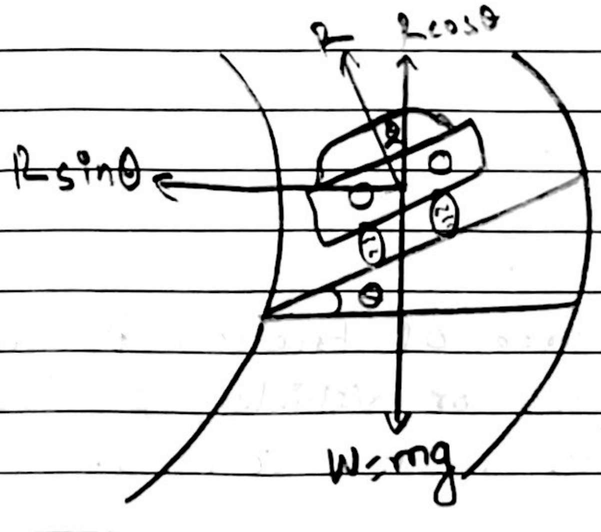
$$\frac{F_f}{R} = \frac{\frac{\cancel{m}v^2}{r}}{\cancel{m}g}$$

or, $\mu = \frac{v^2}{rg}$

$$\boxed{\therefore \mu = \frac{v^2}{rg}}$$

Where μ is coefficient of friction.

- Motion of vehicle on banked circular road



→ The raising of outer edge of road above the inner edge at turning is called banking of road.

Consider a vehicle of mass 'm' is moving round on banked circular road of radius 'r' with uniform speed 'v'. let θ be the angle of banking of the road.

The different forces acting on the vehicle are:-

(i) Its weight (W) = mg, acting vertically downward.

~~Normal~~

(ii) Normal reaction (R)

(iii) frictional force between tyres and road.

Here frictional force is neglected.

The normal reaction (R) can be resolved into two rectangular components $R\sin\theta$ and $R\cos\theta$ as shown in figure.

The component $R\sin\theta$ acts towards the centre of the circular path. so it provides centripetal force to the vehicle.

$$\therefore R\sin\theta = \frac{mv^2}{r} \longrightarrow (1)$$

And,

the component $R\cos\theta$ balances the weight of the vehicle.

$$\therefore R\cos\theta = mg \rightarrow (2)$$

dividing (1) by (2), we get

$$\frac{\cancel{R}\sin\theta}{\cancel{R}\cos\theta} = \frac{\frac{\cancel{m}v^2}{r}}{\cancel{m}g}$$

$$\Rightarrow \tan\theta = \frac{v^2}{rg}$$

$$\Rightarrow v^2 = \tan\theta \, rg$$

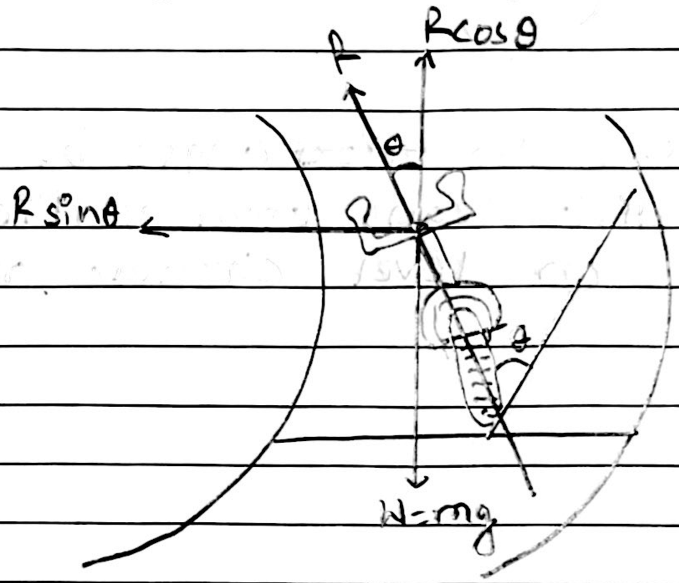
$$\boxed{\therefore v = \sqrt{rg\tan\theta}} \rightarrow (3)$$

This gives maximum speed of vehicle with which we can run the vehicle safely at turning.

Qn: Why the roads are banked at turning?

Ans: If the roads ~~as~~ is not banked at turning then only the frictional force provides centripetal force. if ~~road~~ road is banked at turning then the frictional force ~~is~~ as well as sin component of normal reaction provide centripetal force. As a result, the centripetal force increases and hence vehicle can run safely with greater speed.

- Motion of cyclist on level circular road



Let,

$m \rightarrow$ mass of cycle and cyclist

$\theta \rightarrow$ Angle with which cyclist leans from vertical

$v \rightarrow$ Velocity of cyclist

~~R~~ $r \rightarrow$ radius of circular path.

From figure,

$$R \sin\theta = \frac{mv^2}{r} \longrightarrow (1)$$

$$R \cos\theta = mg \longrightarrow (2)$$

Dividing (1) by (2), we have,

$$\frac{\cancel{R} \sin\theta}{\cancel{R} \cos\theta} = \frac{\frac{\cancel{m}v^2}{r}}{\cancel{m}g}$$

$$\Rightarrow \tan\theta = \frac{v^2}{rg}$$

$$\Rightarrow \boxed{\theta = \tan^{-1}\left(\frac{v^2}{rg}\right)} \quad \text{--- (3)}$$

This gives the ~~velou~~ angle by which the cyclist leans from vertical while moving on level circular road.

* Numerical and Figure Based Questions:

(i) An object of mass 0.5 kg is rotated in a horizontal circle by a string 1 m long. The maximum tension in the string before it breaks is 50 N. What is the greatest number of revolutions per second of the object?

→ Given.

$m = 0.5$ kg

$T = 50$ N

$l = 1$ m

We know,

$$T\cos\theta = mg$$

$$\text{or, } 50\cos\theta = 0.5 \times 10$$

$$\text{or, } \cos\theta = \frac{5}{50}$$

$$\therefore \theta = 84.26^\circ$$

Now,

$$t = 2\pi\sqrt{\frac{l\cos\theta}{g}}$$

$$\text{or, } \frac{1}{f} = 2\pi\sqrt{\frac{1 \times 0.1}{10}}$$

$$\text{or, } \frac{1}{f} = 2\pi(0.1)$$

$$\therefore f = 1.59 \text{ rev/s}$$

② A mass of 1 kg is attached to the lower end of a string 1 m long whose upper end is fixed. The mass is made to rotate in a horizontal circle of radius 60 cm. If the circular speed of the mass is constant, find the tension in the string and the period of motion.

↳ Given,

$m = 1\ kg$

$l = 1 m$

$r = 0.6 m$

Now,

$\sin\theta = \frac{r}{h} = \frac{0.6}{1}$

$= 0.6$

Now,

$\cos\theta = \frac{P}{h} = \frac{\sqrt{h^2 - r^2}}{h} = \frac{\sqrt{1^2 - 0.6^2}}{1}$

Now,

~~Fa~~ $T\cos\theta = mg$

or, $T(0.8) = 1 \times 10$

$\therefore T = 12.5 N$

Again,

$t = 2\pi\sqrt{\frac{L\cos\theta}{g}}$

$= 2\pi\sqrt{\frac{1 \times 0.8}{10}}$

$= 1.77$ sec

3. An object of mass ~~8~~ 8.0 kg is whirled round in a vertical circle of radius 2m with a constant speed of 6 m/s. calculate the maximum and the minimum tensions in the string.

↳ Given,

$m = 8$ kg

$r = 2m$

$v = 6$ m/s

$$T_{max} = \frac{mv^2}{r} + mg$$

or, $$T_{max} = \frac{8 \times 6^2}{2} + 8 \times 10$$

$$= 144 + 80$$

$$T_{max} = 224 \text{ N}$$

Now,

$$T_{min} = \frac{mv^2}{r} - mg$$

$$= \frac{8 \times 6^2}{2} - mg$$

$$= \frac{8 \times 6^2}{2} - 8 \times 10$$

$$= 64 \text{ N}$$

$\therefore T_{min} = 64$ N

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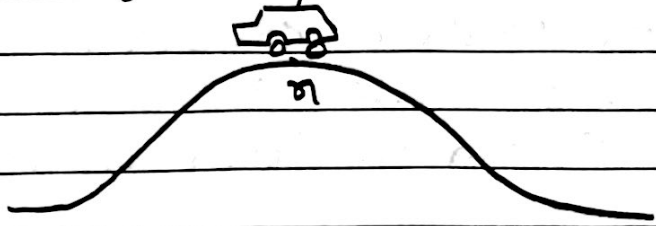
(4) The diagram shows a car of mass 950 kg at the highest point X of the bridge.



The bridge forms part of a verticle circle of radius 20.0 m.
calculate the total upward force of the road on the car:

i) When the car is stationary at x.

→ soln,

$V = 0$.

$r = 20 m$

$m = 950 kg$

Now,

Upward force is given by

$R = mg$

$= 950 \times 9.8$

$= 9310 N$

ii) When the car is passing point X at a speed of 12.0 m/s

→ Here,

$V = 12.0 m/s$

Now,

Upward force = − Tension(T)

$= -\left(\frac{mv^2}{r} - mg\right)$

$$= -\left(\frac{950 \times 12^2}{20} - 950 \times 9.81\right)$$

$$= \cancel{950} \cancel{\times} -(6840 - 9319.5)$$

$$= 2479.5\ N$$

5) A 1400 kg car travelling at 25 m/s rounds a curve of radius 200 m. Find the following.

a) The centripetal acceleration of the car.

→ Given,

$m = 1400\ kg$, $V = 25\ m/s$, $r = 200\ m$

$$a = \frac{v^2}{r}$$

$$= \frac{25^2}{200}$$

$$\therefore a = 3.125\ m/s^2$$

b) The force that maintains centripetal acceleration.

→ $F = ma$

$$= 1400 \times 3.125$$

$$= 4375\ N$$

$$\therefore F = 4375\ N$$

c) The minimum coefficient of friction between the tyres and road that will allow the car to round the curve Safely.

$$\mu = \frac{v^2}{rg}$$

$$= \frac{25^2}{200 \times 10}$$

$$\therefore \mu = 0.3125$$ //.

7) At what angle should a circular road be banked so that a car running at 50 km/hr be safe to go round the circular turn of 200m radius?

→ Given,

$r = 200m$,

$v = 50\ km/hr = \frac{50 \times 1000}{3600}\ m$

$\therefore v = 13.89\ m/s$

We know,

$$v^2 = rg \tan\theta$$

or, $13.89^2 = 200 \times 10 \tan\theta$

or, $192.9 = 2000 \tan\theta$

or, $\tan\theta = 0.0964$

$\therefore \theta = 5.5^\circ$ //.

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⑥ A coin placed on a disc rotates with speed of $33\frac{1}{3}$ rev. min^{-1} provided that the coin is not more than 10 cm. from the axis. Calculate the coefficient of static friction between the coin and the disc.

↳ Soln,

$f = 33\frac{1}{3}$ rev. min^{-1} $= \frac{10\cancel{0}}{3 \times 6\cancel{0}}$ rev/sec

$r = 10\,cm = 0.1\,m$

$\mu = ?$

We know that,

$\mu R = m \times 4\pi^2 f^2 \times r$

$\mu \cancel{m} g = \cancel{m} \times 4\pi^2 f^2 \times r$

$\mu = \frac{4\pi^2 \times \left(\frac{10}{36}\right)^2 \times 0.1}{10}$

$\mu = \frac{4 \times \left(\frac{22}{7}\right)^2 \times \frac{100}{36^2} \times 0.1}{10}$

$= 0.122$

- Conceptual Questions.

① A body is moving with constant velocity v along a circular path of radius r. How much work is done by the centripetal force is revolving the body?

Ans: We know that,

$$W = \frac{1}{2}mv^2 - \frac{1}{2}mu^2$$

For constant velocity.

~~W=~~ $u = v$

$$\text{So, } W = \frac{1}{2}mv^2 - \frac{1}{2}mv^2 = 0$$

or,

In circular motion, the centripetal force acts towards the center of the circle, while the instantaneous displacement is along the tangen. The angle θ between them is always 90°. since cos 90° = 0, the work done is zero.

2) If there is a net force acting on a particle in uniform circular motion, why does the particle's speed not change?

Ans: The net force acting on the particle in circular motion is always perpendicular to its Velocity vector. So A force acting perpendicular to motion cannot perform work or change the direction

of the velocity vector.
According to work-Energy theorem,
Since no work is done,

$$W = \frac{1}{2}mv^2 - \frac{1}{2}mu^2$$

$$0 = \frac{1}{2}mv^2 - \frac{1}{2}mu^2$$

$$\frac{1}{\cancel{2}}\cancel{m}v^2 = \frac{1}{\cancel{2}}\cancel{m}u^2$$

$$\therefore v = u$$

③ Why does a cyclist lean from vertical while turning on curved track?

Ans: A cyclist leans himself to the vertical while moving round a circular path to provide necessary centripetal force. The normal reaction of the road makes an angle with the vertical and horizontal component of the reaction provides the necessary centripetal force to move the cycle in the circular path.

④ A stone tied at one end of a string is whirled in a vertical circle. Where will the tension in the string be maximum and minimum? Show them in a diagram.

Ans: Net force on particle towards the centre is given by

$T = mg\cos\theta$

This provides centripetal force Then,

$T = mg\cos\theta$

$= \frac{mv^2}{r}$

At lower point (A)

$F = \frac{mv^2}{r} + mg$ (max)

At high point (B)

$T = \frac{mv^2}{r} - mg\cos 180^\circ$

$T = \frac{mv^2}{r} - mg$ (min)

Hence tension is maximum at A and lowest at B.